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ABSTRACT

The Online Shopping Cart System is a comprehensive software application designed to provide a digital marketplace experience within a desktop environment. Traditionally, shopping involved physical presence, manual inventory tracking, and paper based billing, which was time-consuming and prone to human errors. This project provides a robust solution developed using Java Swing for the user interface and Oracle Database for persistent storage. The application facilitates core retail functions including user authentication, real-time product browsing, dynamic cart management, and automated invoice generation. By utilizing JDBC(Java Database Connectivity), the system ensures that inventory levels are updated instantly upon purchase. This report details the full software development life cycle (SDLC) of the system, including requirement analysis, database schema design, system architecture, and final implementation results.

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ACRONYMS

- **SRMS** : Online Shopping Card System
- **JDBC** : Java Database Connectivity
- **SQL** : Structured Query Language
- **DBMS** : Database Management System
- **GUI** : Swing Based

1.INTRODUCTION

1.1 Overview

The retail industry has undergone a massive transformation with the advent of digital technology. An Online Shopping Cart system is the backbone of any e-commerce platform. It allows users to select products, review their choices, and proceed to a checkout process. While most modern systems are web-based, this project focuses on a high-performance desktop application model using Java, which is ideal for dedicated kiosks or internal retail management systems The system is developed using Core Java and JDBC, with MySQL as the backend database. It allows administrators to manage records and students to view their results securely.

1.2 Problem Statement

Manual management of shopping processes leads to:

- Multiple entries for the same product causing confusion.
- Calculation of bills and inventory updates manually takes time.
- Lack of authorized access to sensitive sales data.

1.3 Motivation and Contribution

The primary motivation for this project is to build a secure, efficient, and user-friendly platform that bridges the gap between manual shopping and complex cloud-based systems To develop a system to manage student and result data

1. Using Java Swing to create an intuitive interface.
2. Implementing Oracle constraints to prevent invalid data entry.
3. Automated total calculation and invoice generation.

1.4 Obejectives

- To design a secure login module for users and administrators.
- To implement a searchable product catalog.
- To develop a ”Cart” logic that allows adding, removing, and updating product quantities.
- To ensure seamless connectivity with an Oracle 19c/21c database.
- To generate a final bill with calculated taxes and discounts.

1.5 Scope of the Project

The scope includes the development of the frontend using Java (AWT/Swing), back- end logic using Java SE, and database management using Oracle SQL. The applica- tion is designed to run on a local network (LAN) environment or as a standalone retail terminal.

2. SYSTEM ANALYSIS

2.1 Feasibility Study

Before proceeding with development, a feasibility study was conducted:

1. Technical Feasibility: Java and Oracle are mature technologies with extensive documentation. The developer has sufficient expertise to implement JDBC.
2. Operational Feasibility: The system requires minimal training for users who have basic computer knowledge.
3. Economic Feasibility: The project uses standard hardware and open-source or community editions of software, making it cost-effective.

2.2 Requirements Specification

This section specifies the hardware and software needed to run the system.

2.2.1 Functional Requirements

- User Authentication: Users must sign in to access the cart.
- Product Management: Admin can add, update, or delete products.
- Search Functionality: Users can search products by name or category.
- Cart Operations: Add to cart, view total, and clear cart functionality.
- Checkout: Transactional commitment of the purchase to the database.

2.2.2 Software and Hardware Requirements

2.3.1 Hardware Requirements

- Processor: Intel Core i3 or higher / AMD Ryzen 3 or higher.
- RAM: Minimum 4 GB (8 GB recommended for Oracle Database).
- Storage: 500 MB for application and 2 GB for database files.
- Display: 1366x768 resolution or higher.

2.3.1 Software Requirements

- **Operating System:** Windows 10/11 or Linux.
- **Language:** Java Development Kit (JDK) 17 or higher.
- **IDE:** Visual Studio Code or IntelliJ IDEA.
- **Database:** Oracle Express Edition (XE).
- **Driver:** Oracle JDBC Thin Driver (ojdbc8.jar).

3. TECHNOLOGY DESCRIPTION

3.1 Java Technology

Java was chosen for its platform independence (Write Once, Run Anywhere)

3.1.1 Java Swing

Swing is a GUI widget toolkit for Java. It is part of Oracle's Java Foundation Classes (JFC). It provides a more sophisticated set of GUI components than the earlier Abstract Window Toolkit (AWT).

3.2.1 JDBC (Java Database Connectivity)

JDBC is an API that allows Java applications to interact with various databases. It provides methods for querying and updating data in a relational database.

3.2 Orcale Database

Oracle is a multi-model database management system. It is widely used for running online transaction processing (OLTP). In this project, it provides high data integrity and security features.

4. DATABASE DESIGN

4.1Enity Relationship (ER) Diagram

The system consists of three primary entities: Users, Products, and Orders.

- A User can place many Orders.
- An Order can contain many Products (Many-to-Many).

4.2 Data Dictionary

4.2.1: User Table Structure

Field Name	Data Type	Constraint	Description
User_ID	Number	Primary Key	Unique ID for each user
Username	Varchar2(50)	Not Null	Login name
Password	Varchar2(50)	Not Null	Encrypted password
Email	Varchar2(100)	Unique	User contact

4.2.2: Product Table Structure

Field Name	Data Type	Constraint	Description
Prod_ID	Number	Primary Key	Unique product ID
Prod_Name	Varchar2(100)	Not Null	Name of product
Price	Number(10,2)	Check > 0	Unit price
Stock	Number	Default 0	Available quantity

5. SYSTEM DESIGN and Architecture

5.1 System Architecture

The project follows a **Two-Tier Architecture**:

1. **Client Tier (Frontend):** The Java Swing application handles user interaction and business logic.
2. **Database Tier (Backend):** The Oracle Database handles data persistence, constraints, and triggers.

5.2 Modules Description

1.Authentication Module:

- Handles login and registration.

2.Catalog Module:

- Fetches product data from the database and displays it in a JTable.

3.Cart Module:

- A temporary data structure (ArrayList) that stores selected items before final checkout.

4.Billing Module:

- Calculates the final amount and updates the Oracle ‘stock‘ col- umn using SQL UPDATE queries

6. SYSTEM IMPLEMENTATION

6.1 Database Connectivity Implementation

Connection to Oracle is established using:

```
DriverManager.getConnection("jdbc:oracle:thin:@ "system",  "password");
```

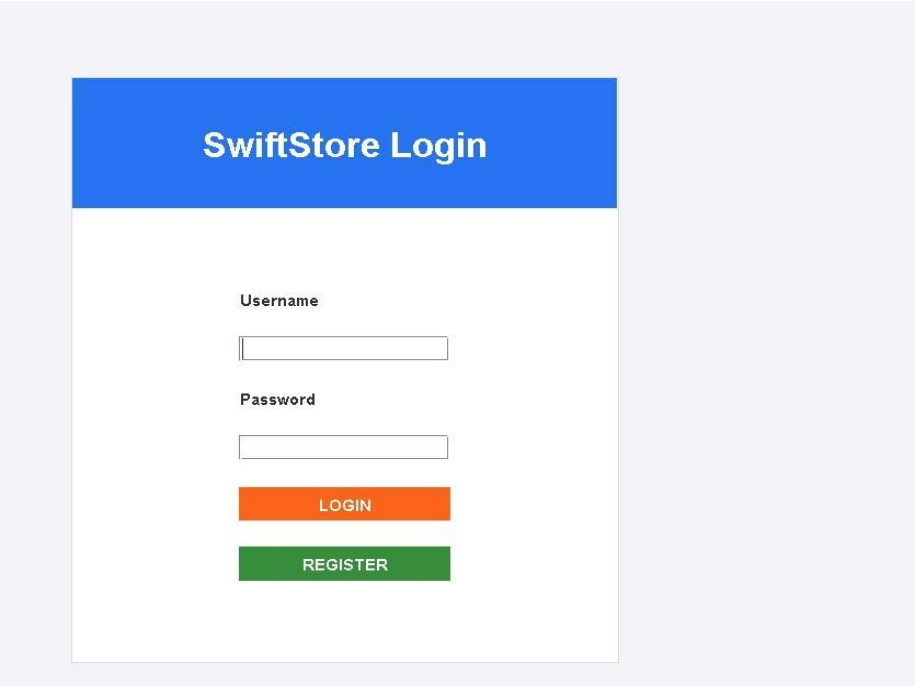

Figure 6.1: Login Screen of Online Shopping Cart System

Figure 6.2: New user account creation interface

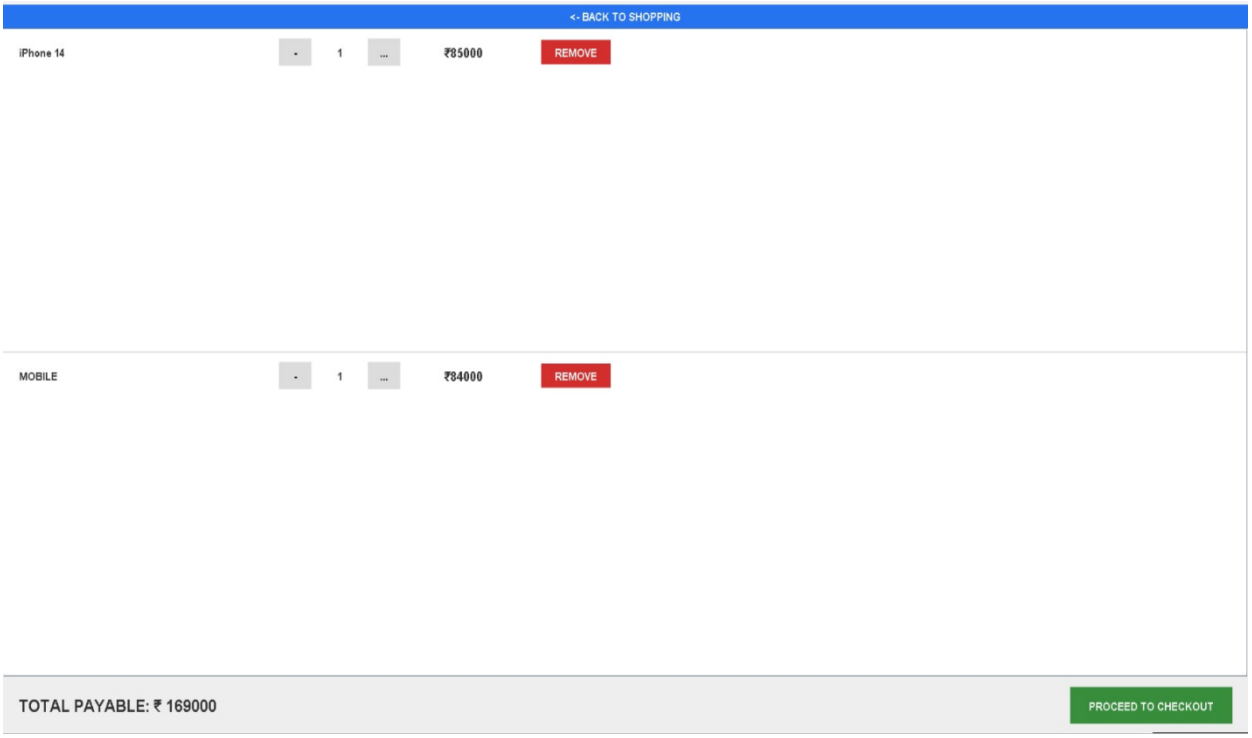

Figure 6.3: Main product catalog after login

```java
private JPanel createDashboardScreen() {
    JPanel mainDash = new JPanel(new BorderLayout());

    JPanel header = new JPanel(new BorderLayout());
    header.setBackground(PRIMARY_BLUE);
    header.setPreferredSize(new Dimension(width: 1200, height: 65));

    JLabel titleLabel = new JLabel(text: "SwiftStore Pro", JLabel.CENTER);
    titleLabel.setForeground(Color.WHITE);
    titleLabel.setFont(new Font(name: "Segoe UI", Font.BOLD, size: 22));
    header.add(titleLabel, BorderLayout.CENTER);

    JButton logout = createStyledButton(text: "<- Logout", Color.WHITE, PRIMARY_BLUE);
    logout.addActionListener(e -> {
        sessionUserId = -1;
        sessionUsername = "";
        cartItems.clear();
        updateCartCount();
        cardLayout.show(mainPanel, name: "LOGIN");
```

Figure 6.4 Backend and Logic Implementyation

```
mysql> desc Order_items;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| id           | int          | NO   | PRI | NULL    | auto_increment |
| order_id     | varchar(20)  | YES  |     | NULL    |                |
| product_name | varchar(200) | YES  |     | NULL    |                |
| quantity     | int          | YES  |     | NULL    |                |
| price        | double       | YES  |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
5 rows in set (0.161 sec)
```

```
mysql> use e_commerce
Database changed
mysql> show tables
    -> ;
+----------------------+
| Tables_in_e_commerce |
+----------------------+
| order_items          |
| orders               |
| products             |
| users                |
+----------------------+
4 rows in set (0.557 sec)
```

database > DBConnection.java

```java
package database;
import java.sql.Connection;
import java.sql.*;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
public class DBConnection {
    private static final String url = "jdbc:mysql://localhost:3306/e_commerce";
    private static final String username = "root";
    private static final String password = "Dadzadee@06";
    static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
    public static Connection getConnection() {
        Connection conn = null;
        try {
            conn = DriverManager.getConnection(url, username, password);
            System.out.println("Database connected successfully");
        } catch (SQLException e) {
            System.out.println("Database connection failed");
            e.printStackTrace();
        }
        return conn;
```

Unable to determine the current SDK. Use the SDK Manager to set the c...

```
mysql> desc orders;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| order_id      | varchar(20) | NO   | PRI | NULL    |       |
| user_id       | int         | YES  |     | NULL    |       |
| total         | double      | YES  |     | NULL    |       |
| status        | varchar(20) | YES  |     | NULL    |       |
| order_date    | datetime    | YES  |     | NULL    |       |
| delivery_date | varchar(50) | YES  |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
6 rows in set (0.023 sec)
```

```
mysql> desc users;
+----------+--------------+------+-----+---------+----------------+
| Field    | Type         | Null | Key | Default | Extra          |
+----------+--------------+------+-----+---------+----------------+
| ID       | int          | NO   | PRI | NULL    | auto_increment |
| USERNAME | varchar(255) | YES  | UNI | NULL    |                |
| EMAIL    | varchar(255) | YES  |     | NULL    |                |
| PASSWORD | varchar(255) | YES  |     | NULL    |                |
| PHONE    | varchar(255) | YES  |     | NULL    |                |
+----------+--------------+------+-----+---------+----------------+
5 rows in set (0.030 sec)
```

```
mysql> select *from orders;
+----------+---------+--------+-----------+---------------------+---------------+
| order_id | user_id | total  | status    | order_date          | delivery_date |
+----------+---------+--------+-----------+---------------------+---------------+
| ORD17963 |      12 | 163000 | Cancelled | 2026-03-28 10:34:57 | 2026-04-02    |
| ORD25140 |      12 | 169000 | Cancelled | 2026-03-29 10:54:09 | 2026-04-02    |
| ORD35381 |      10 |  78000 | Ordered   | 2026-03-17 13:36:49 | 2026-03-20    |
| ORD44069 |      11 | 253000 | Ordered   | 2026-03-23 12:22:21 | 2026-03-28    |
| ORD59285 |      11 |  84000 | Ordered   | 2026-03-23 12:21:42 | 2026-03-28    |
+----------+---------+--------+-----------+---------------------+---------------+
5 rows in set (0.309 sec)
```

```
mysql> select *from products;
+----+--------------------+-------------+-------+-----------------------+--------+---------+
| id | name               | category    | price | specs                 | rating | brand   |
+----+--------------------+-------------+-------+-----------------------+--------+---------+
| 1  | MOBILE             | 14 PRO MAX  | 84000 | 16 GB RAM | 128 GB RAM |    3.5 | IPHONE  |
| P1 | iPhone 14          | Mobiles     | 85000 | 128GB Storage         |    4.5 | Apple   |
| P2 | Samsung Galaxy S23 | Mobiles     | 78000 | 256GB Storage         |    4.4 | Samsung |
| P3 | Laptop Pro         | Electronics | 65000 | 16GB RAM              |    4.3 | Dell    |
+----+--------------------+-------------+-------+-----------------------+--------+---------+
4 rows in set (0.115 sec)
```

```
mysql> select *from users;
+----+----------+-------------------+------------------------------------------------------------------+---------+
| ID | USERNAME | EMAIL             | PASSWORD                                                         | PHONE   |
+----+----------+-------------------+------------------------------------------------------------------+---------+
|  3 | dadza    | Dadzadeee@06      | 98665                                                            |         |
|  5 | moyo     | moyo@gmail.com    | dadzadee@06                                                      | 88888   |
|  6 | chirandu | chirandu@gmail.com | dadzadee06                                                       | 988888  |
|  7 | abc      | abc@gmail.com     | abcd                                                             | 12345   |
|  8 | dc       | dc@gmail.com      | dc                                                               | 123     |
|  9 | 123      | 1232@gmail.com    | 12345                                                            | 123456  |
| 10 | robbo26  | robo@gmail.com    | 047066c004315b981ff3f7fac0da4523d49bf0bbf7697fb4bc377a2bc81296e4 | 12345   |
| 11 | vamoyo   | r@gmail.com       | f29ad3f66693773190f34911777ea42c0e9523a3fc697ff8bb4e92725e196b98 | 12345   |
| 12 | theo     | theo@gmail.com    | 1cf8531a22348eeb2abe9b891c939707aa5abfab759f1f6be7abc4a763968ebc | 12345   |
+----+----------+-------------------+------------------------------------------------------------------+---------+
9 rows in set (0.091 sec)
```

7. CONCLUSION

7.1 Conclusion

The Online Shopping Cart System was successfully developed. The integration be- tween Java Swing and Oracle proved to be highly efficient for desktop-based retail operations. The system meets all defined functional requirements and provides a secure environment for transactions.

8.FUTURE ENHANCEMENTS

1.Web Integration:

- Transitioning the frontend to a web-based React or Angular framework.

2.Mobile App:

- Developing a companion Android/iOS app.

3.Payment Gateway:

- Integrating real-world APIs like Stripe or PayPal.

4.AI Recommendations:

- Adding a suggestion engine based on user purchase history.

9. REFERENCES

The following references were used during the development of the Employee Management System:

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   https://docs.oracle.com
5. **Java Documentation (Oracle)**,
   https://docs.oracle.com/javase
6. **JDBC Tutorial – GeeksforGeeks**,
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7. **Java Swing Tutorial – TutorialsPoint**,
   https://www.tutorialspoint.com/java_swing
8. **W3Schools Java Tutorial**,
   https://www.w3schools.com/java/

APPENDIXA

A.1 Database Connection Class

```
import java.sql.*;

public class DBConnection {

    public static Connection getConnection() {
        Connection con = null;

        try {

            Class.forName("oracle.jdbc.driver.OracleD
            river"); con =
            DriverManager.getConnection(

                "jdbc:oracle:thin:@localhost:1521:xe","system","password");

        } catch (Exception e) {
            System.out.println("Connection Failed: "
            + e);

        }

        return con;

        }

}
```

A.2 SQL Schema

```
CREATE TABLE products (

    prod_id NUMBER PRIMARY KEY,

        prod_name VARCHAR2(100), price NUMBER(10,2)

    stock NUMBER

);


INSERT INTO products VALUES (1, 'Laptop', 75000, 10);

INSERT INTO products VALUES (2, 'Smartphone', 25000, 50);
```